

Leyan Li

leyanpower@gmail.com / 3073751449@qq.com | +86 18805506898 | GitHub: <https://github.com/lionlepower>

Education

University of Edinburgh, United Kingdom | Sep 2025 - Sep 2026 Expected

MSc High Performance Computing with Data Science

University of Liverpool, United Kingdom | Sep 2023 - Jun 2025

BSc Computer Science, Xi'an Jiaotong-Liverpool University 2+2 Programme, First Class Honours

Technical Skills

Programming & Engineering: Python, C/C++, SQL, Bash; developed backend APIs, data pipelines, log parsers, experiment automation scripts, model-training workflows, and performance-analysis utilities; basic JavaScript / HTML / CSS for web prototypes.

AI Agent & LLM Applications: Implemented agent workflows with tool selection, task decomposition, RAG, prompt templates, multi-turn state, short-term memory, Markdown report generation, and integrations with files, databases, caches, and automation scripts.

RAG & Retrieval: Built Markdown ingestion pipelines with chunking, embeddings, FAISS / Chroma vector indexes, top-k semantic retrieval, context construction, and retrieval-augmented QA / reporting workflows.

Backend & System Design: FastAPI, REST APIs, HTTP, request validation, exception handling, logging, TraceID-based debugging, task scheduling, rate limiting, idempotency, and observability concepts.

Databases & Caching: MySQL / SQLite schema design, indexing, transactions, locks, and query optimization; Redis hot-query caching, session state, TTL lifecycle control, cache penetration / breakdown mitigation, and consistency trade-offs.

ML & Data Analysis: PyTorch, multi-agent reinforcement learning experiments, Pandas, NumPy, Matplotlib, structured log analysis, metric calculation, and visualization.

Tools: Linux, Git, VS Code, Docker basics, Slurm, PETSc, MPI, OpenMP, reproducible documentation, and script-driven experiment management.

Projects

AI Agent System for HPC Benchmark Performance Analysis

Tech Stack: Python, FastAPI, LangChain / LlamaIndex, FAISS / Chroma, OpenAI API / Local LLM, SQLite / MySQL, Redis, Pandas, Matplotlib, PETSc, MPI, OpenMP, Slurm, Docker, Linux

Project Scope: Developed an AI Agent system for PETSc benchmark management, Slurm / PETSc log analysis, historical result querying, HPC knowledge retrieval, and Markdown performance report generation. The system supports natural-language queries over real 2D / 3D stencil benchmark data and compares MPI-only vs hybrid MPI+OpenMP configurations.

- Architected an 8-layer agent system covering API services, planning, tool execution, RAG retrieval, relational storage, Redis caching, report generation, and observability for multi-step benchmark analysis workflows.
- Delivered FastAPI endpoints for natural-language QA, log upload, experiment querying, history lookup, metric calculation, chart generation, and Markdown report export, with unified validation, exception handling, and response formatting.
- Orchestrated a tool-routing workflow that selects among database query, Slurm / PETSc log extraction, metric computation, knowledge retrieval, visualization, and report-generation tools according to the user request.
- Indexed PETSc, MPI, OpenMP, Slurm, Linaro MAP, and personal HPC notes through Markdown parsing, chunking, embedding generation, FAISS / Chroma indexing, and top-k semantic retrieval for grounded performance explanations.
- Modelled benchmark records in SQLite / MySQL, storing configurations, execution status, runtime metrics, error logs, and analysis history; enabled filtering by problem size, execution mode, total cores, and creation time.
- Optimized common query paths with indexes for best-configuration lookup, MPI-only vs MPI+OpenMP comparison, and time-range experiment history; added Redis TTL caching for hot queries, session state, and query-hash analysis results.
- Converted raw Slurm / PETSc outputs into structured records by extracting problem size, nodes, MPI ranks, OpenMP threads, total cores, runtime, execution mode, solver status, and error messages.
- Automated runtime, speedup, and parallel-efficiency calculations with Pandas and Matplotlib, producing comparison charts across problem sizes and parallel configurations for Markdown reports.

- Instrumented each agent request with TraceID logging for user questions, tool chains, retrieved documents, cache hits, database latency, RAG retrieval output, and LLM latency to debug retrieval failures and API bottlenecks.
- Validated the workflow on real PETSc 2D / 3D stencil benchmarks, including 40,960,000-unknown 2D and 41,063,625-unknown 3D cases under MPI-only and hybrid MPI+OpenMP runs with 1-2 nodes, 16 MPI ranks per node, and 8 OpenMP threads per rank.

Value Function Decomposition in Multi-Agent Actor-Critic Methods

Tech Stack: Python, PyTorch, ePyMARL, PyMARL, MAPPO, VDN, QMIX, Multi-Agent Reinforcement Learning

Project Scope: Researched factorised critics in MAPPO by integrating VDN and QMIX-style value decomposition to study cooperation efficiency, convergence stability, and policy performance in cooperative multi-agent environments.

- Implemented 3 algorithm variants, MAPPO, MAPPO-VDN, and MAPPO-QMIX, on ePyMARL / PyMARL, covering training configuration, experiment execution, evaluation, and result analysis.
- Reworked the critic architecture with VDN additive decomposition and QMIX monotonic mixing networks to test factorised value estimation inside actor-critic training.
- Evaluated algorithms in 2 cooperative environments, Matrix Game and Predator-Prey, using test return, convergence speed, final return, cooperation quality, and training stability.
- Produced reward-curve visualizations and diagnostic analysis for instability sources, including critic approximation error, multi-agent non-stationarity, credit assignment, and mixing-network representation limits.
- Authored a full academic paper covering algorithm design, implementation details, experimental results, limitations, and future work.

Publication

Li, Leyan. Convolutional Neural Networks Based Medical Image Analysis. International Conference on Engineering Management, Information Technology and Intelligence, EMITI 2024.